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DIRECTORATE OF ONLINE EDUCATION

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

KATTANKULATHUR - 603 203

MCA III SEM

MINI PROJECT VIVA VOCE

NOV 2026

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PROJECT OVERVIEW

TITLE OF MINI PROJECT

MCA III Semester Team / Individual Software Engineering Project

TITLE OF MINI PROJECT / INTERNSHIP

COMMUNITY ITEM LENDING SYSTEM

A Secure Web Platform for Hyperlocal Neighborhood Resource Sharing & Circular Economy

PROBLEM DOMAIN

Hyperlocal Collaborative Consumption

- **Household Overspending:** Prevents duplicate buying of costly tools.
- **Domestic Clutter:** Shares items that are needed only rarely.
- **Community Trust:** Connects verified residential neighbors safely.

TECHNOLOGY STACK

Full-Stack Web Architecture

- **Backend Framework:** Python 3.12 + Flask 3.0 WSGI Engine.
- **Cloud Database:** TiDB Cloud Serverless (MySQL 8.0 Protocol).
- **Frontend Design:** Responsive HTML5, CSS3, Bootstrap 5.3.

EXPECTED IMPACT

Economic & Environmental Benefits

- **Financial Savings:** Zero rental fees; eliminates deposit barriers.
- **Carbon Reduction:** Cuts manufacturing demand for duplicate goods.
- **Circular Economy:** Extends lifespan of existing community resources.

EXECUTIVE SUMMARY

ABSTRACT

What the Project Solves and Delivers for Community Residents

THE PROBLEM

Redundant Domestic Purchases & High Rental Costs

- **Household Underutilization:** Common tools like drills and ladders are used for less than 15 minutes a year, then sit idle in storage.
- **Heavy Financial Outlay:** Buying infrequently used appliances wastes household budgets and clutters living spaces unnecessarily.
- **Commercial Rental Burdens:** Rental shops charge high daily fees, transport costs, and demanding security deposits that deter borrowing.

THE SOLUTION

Automated Peer-to-Peer Neighborhood Sharing

- **Centralized Digital Catalog:** Verified residents easily list spare tools, books, and appliances with photos and condition details.
- **One-Click Borrowing:** Borrowers search local items, check live availability, and send instant requests with desired return dates.
- **State Machine Workflow:** Automated 4-stage tracking: Available -> Pending -> Lent -> Returned, preventing double-booking.

EXPECTED VALUE

Community Cohesion & Environmental Sustainability

- **Financial Savings:** Neighbors save thousands of rupees each year through free community borrowing without rental fees.
- **Eco-Friendly Circular Economy:** Cuts duplicate manufacturing demand, reducing plastic, electronic, and metal waste.
- **Mutual Neighborhood Trust:** Encourages communication, accountability, and goodwill between residential community members.

CORE CAPABILITIES

KEY FEATURES

Essential Functionality Built for Neighborhood Communities

[01] Easy Item Cataloging

List surplus household tools in seconds with clear photos, descriptions, and category tags.

- **Photo Upload with UUIDs:** Secure file sanitization ensures unique, uncollided image names.
- **Structured Categories:** Tools, Gardening, Kitchen, Books, Electronics, and Sports.
- **Condition Guidelines:** Allows owners to describe item condition and usage tips.

[02] Smart Search & Filtering

Find needed equipment within the neighborhood immediately without scrolling through chats.

- **Keyword Query Engine:** Search by item title or description in real time.
- **One-Click Category Filters:** Interactive badge pills to narrow down available listings.
- **Real-Time Status Badges:** Instant visual badges for Available, Pending, and Lent items.

[03] Automated Lending Workflow

Streamlined 4-stage lifecycle to ensure smooth borrowing without misunderstandings.

- **Digital Borrow Requests:** Select return dates and include custom messages to the owner.
- **Owner Approval Controls:** One-click approval or rejection right from the user dashboard.
- **Automatic Concurrency Lock:** Items automatically lock to prevent double-booking.

[04] Audit History & Profile Dashboard

Complete accountability and transparency for all community transactions.

- **Unified Dashboard:** Three tabs for Listed Items, Received Requests, and Active Borrows.
- **Permanent Audit Trail:** Records timestamps, borrower names, and return dates.
- **Cryptographic Security:** PBKDF2:SHA256 password hashing and secure session control.

INTRODUCTION & PROBLEM CONTEXT

Why Neighborhood Resource Sharing Matters and How We Solve It

THE DOMESTIC TOOL PARADOX

- **12-Minute Lifespan:** Studies show the average power drill is operated for only 12 to 15 minutes during its entire working life.
- **Redundant Purchases:** In an apartment complex of 100 families, dozens of identical lawn mowers, ladders, and tools are bought redundantly.
- **Storage Clutter & Waste:** Items sit idle in cupboards for years, losing value while creating domestic clutter and electronic waste.
- **Existing Channels Fall Short:** Retail sites only push new purchases; classifieds focus on disposal; rental shops charge expensive daily fees.

OUR AIM & AUTOMATED SOLUTION

- **Centralized Sharing Portal:** Develop a lightweight, secure web app where community members list and discover idle household items.
- **Structured Borrowing Workflow:** Digital borrow requests with dates and messages replace messy WhatsApp messages and verbal requests.
- **Zero-Rental Barrier:** Allows free community lending between trusted neighbors, saving money and building neighborhood cohesion.
- **Automated Tracking & Audit:** 4-stage state machine (Available -> Pending -> Lent -> Returned) ensures complete accountability.

TECHNICAL SPECIFICATIONS

SYSTEM REQUIREMENTS

Hardware and Software Environment Specifications

HARDWARE REQUIREMENTS

Component	Requirement Specification
Processor	Intel Core i3 / i5 or AMD Ryzen 3 / 5 (2.0 GHz+)
RAM	4 GB Minimum (8 GB Recommended for multi-tasking)
Hard Disk	10 GB Available free storage for code and uploaded media
Display Monitor	1366 x 768 or Full HD 1920 x 1080 resolution
Input Devices	Standard Keyboard and Optical Mouse / Trackpad
Network	Broadband / 4G Internet connectivity for TiDB Cloud

SOFTWARE REQUIREMENTS

Component	Selected Technology
Operating System	Microsoft Windows 10 / 11, Linux, or macOS
Programming Language	Python 3.10 / 3.12 (Standard Runtime)
Web Framework	Flask 3.0+ (Microframework with Jinja2 Templating)
Database Engine	TiDB Cloud Serverless (MySQL 8.0 Protocol with TLS)
Frontend Technologies	HTML5, CSS3, JavaScript (ES6), Bootstrap 5.3
QA & Tools	Pytest, VS Code, Git, GitHub Version Control

COMPARATIVE ANALYSIS

EXISTING SYSTEM VS. PROPOSED SYSTEM

Contrasting Traditional Informal Borrowing with Our Automated Solution

EXISTING INFORMAL METHODS (GAPS)

- **Zero Catalog Visibility:** Neighbors have no directory to know what tools others own, leading to awkward or unanswered chat requests.
- **Lack of Accountability:** Informal verbal agreements result in forgotten return dates, lost items, and untracked damages.
- **Double-Booking Conflicts:** Without a centralized schedule, multiple borrowers can request the same item at the same time.
- **High Financial Burden:** Residents spend thousands of rupees buying tools they only need for one quick home repair.

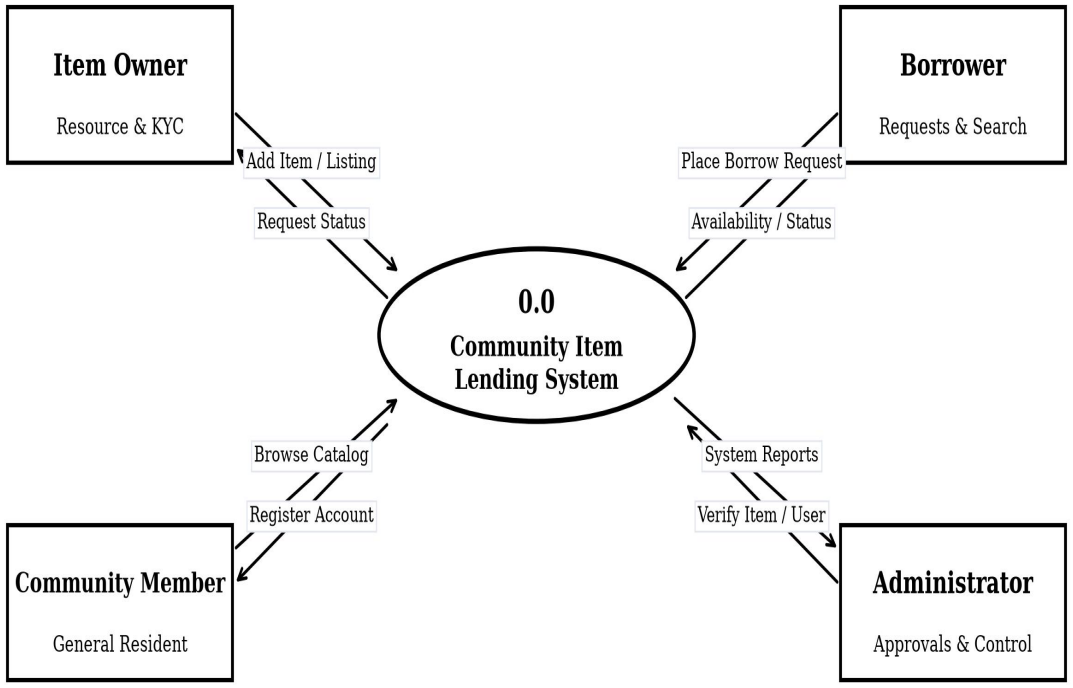
PROPOSED AUTOMATED PLATFORM (SOLUTIONS)

- **Centralized Online Catalog:** Searchable digital directory of all available items with photos, condition notes, and categories.
- **Automated 4-Stage State Machine:** Lifecycle shifts automatically: Available -> Pending -> Lent -> Returned, preventing scheduling clashes.
- **Permanent Audit Trail:** Every transaction is permanently logged with timestamps, borrower details, approvals, and return records.
- **Zero-Cost Circular Sharing:** Saves household funds, eliminates rental deposits, and reduces duplicate manufacturing waste.

SYSTEM DESIGN - DATA FLOW DIAGRAM

Information Flows Across Users, Application Logic, and Cloud Database

Figure 1: DFD Level 0 – Context Diagram



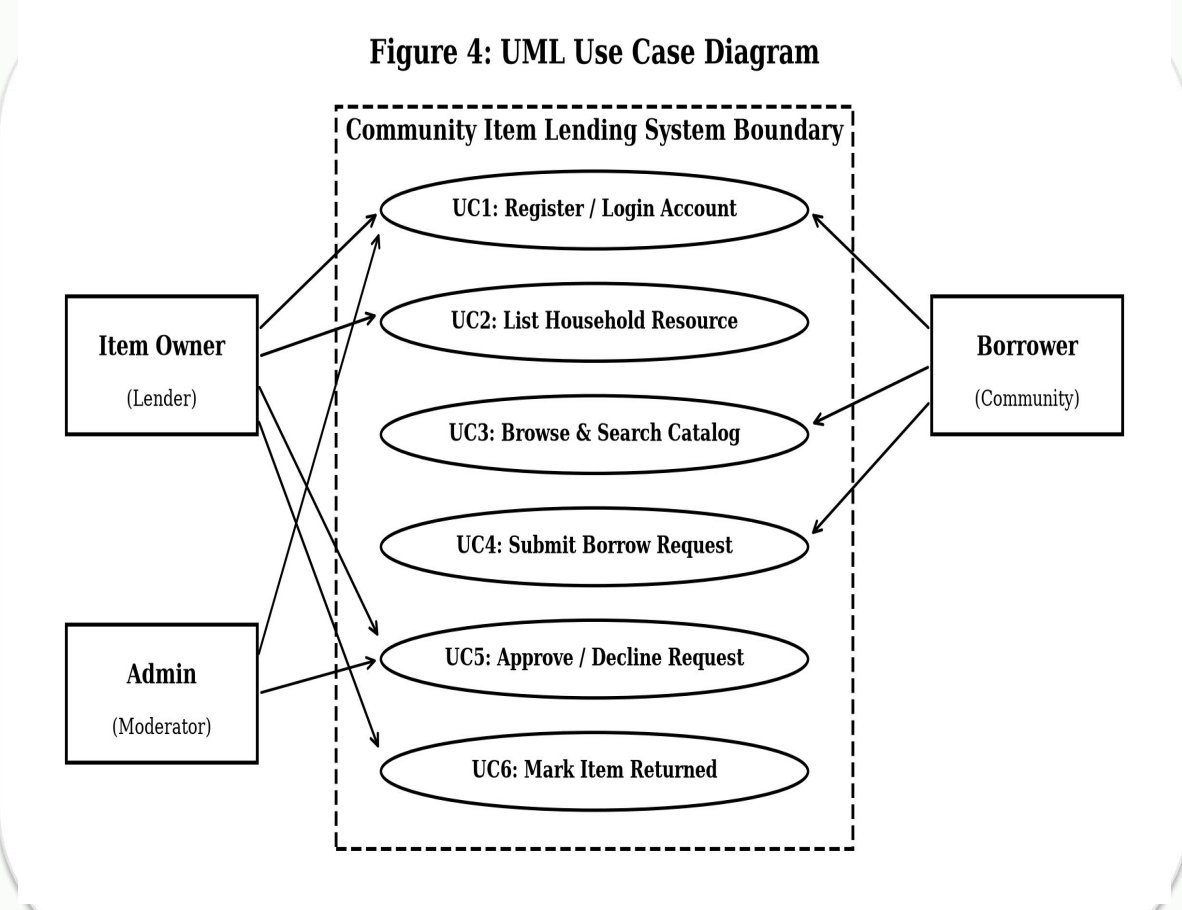
CORE DATA FLOW STAGES

- **1. User Registration & Auth:** Credentials and room details sent via HTTPS; passwords encrypted with PBKDF2:SHA256 and stored in users table.
- **2. Item Upload & Cataloging:** Lender submits title, category, and photo; backend assigns UUID filename and saves metadata in items table.
- **3. Borrow Request Workflow:** Borrower requests an item; record inserted into borrow_requests; item status transitions to Pending.
- **4. Handover & Return Confirmation:** Owner approves request (status -> Lent); upon return, owner confirms handover, resetting item to Available.

INTERACTION MODEL

SYSTEM DESIGN - USE CASE DIAGRAM

Actor Roles and Functional System Boundaries



ACTOR RESPONSIBILITIES

Item Owner (Lender):

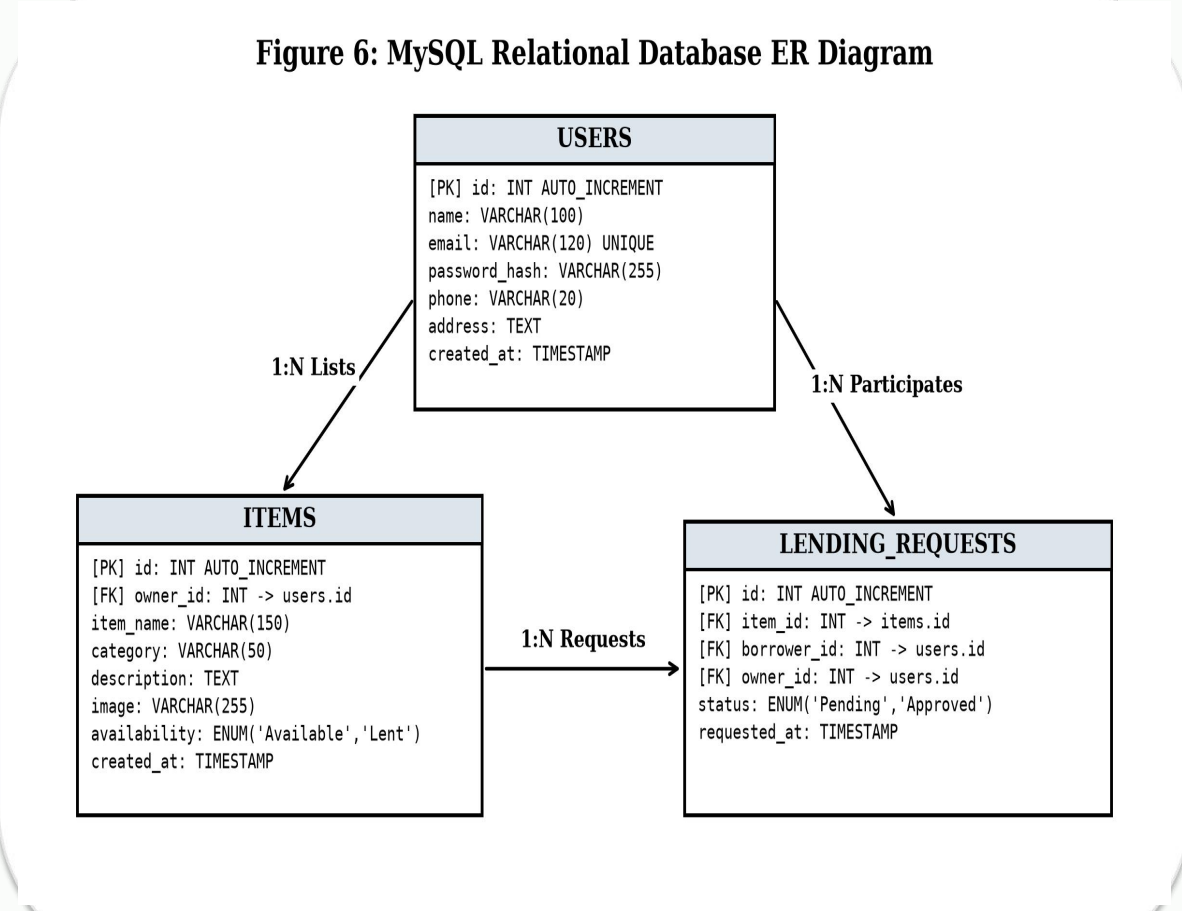
- **Add New Item:** List spare household items with title, category, description, and photo.
- **Manage My Listings:** View, edit, or remove listed items anytime.
- **Review Incoming Requests:** Approve or decline borrowing requests from neighbors.
- **Confirm Return:** Verify item receipt and reset status to Available.

Community Borrower:

- **Search & Browse:** Explore catalog by keyword search and category filters.
- **Submit Request:** Request item specifying duration and purpose note.
- **Track Status:** Monitor approvals and active loans in dashboard.

DATABASE DESIGN - ENTITY-RELATIONSHIP MODEL

Normalized Relational Schema on TiDB Cloud Serverless (MySQL 8.0 Protocol)



RELATIONAL SCHEMA ENTITIES

- **Table: users:** id (PK, Auto-inc), username (VARCHAR 50), email (VARCHAR 100, Unique), password_hash (VARCHAR 255), phone, room_number, created_at.
- **Table: items:** id (PK, Auto-inc), title (VARCHAR 100), description (TEXT), category (VARCHAR 50), image_url (VARCHAR 255), owner_id (FK -> users.id), status (Available/Pending/Lent), created_at.
- **Table: borrow_requests:** id (PK, Auto-inc), item_id (FK -> items.id), borrower_id (FK -> users.id), request_date, return_date, status (Pending/Approved/Rejected/Returned).
- **Referential Integrity:** ON DELETE CASCADE on borrow_requests ensures clean database cleanup if items or users are removed.

MODULE DESCRIPTION - OVERVIEW

Summary of the Six Core Functional Modules

Module ID	Module Name	Primary Functional Responsibility	Underlying Technology
MOD-01	**User Authentication & Security**	Registration, Login, PBKDF2 Password Hashing, Session Management	Flask Sessions & Werkzeug
MOD-02	**Item Catalog & Photo Upload**	Listing surplus items, category tagging, secure UUID image upload	UUID4 & Pillow Sanitization
MOD-03	**Search & Dynamic Filtering**	Real-time search queries, category badge filters, status filters	SQL LIKE Queries & Jinja2
MOD-04	**Borrowing & Lending Workflow**	Request submission, owner approval/rejection, state transitions	Atomic State Machine Logic
MOD-05	**Return Confirmation & Audit**	Handover confirmation, reset to Available, permanent audit history	Transaction Logs in TiDB
MOD-06	**User Profile & Dashboard**	Unified dashboard: Listed Items, Incoming Requests, Active Borrows	Bootstrap 5.3 Responsive UI

IMPLEMENTATION EVIDENCE

SAMPLE SOURCE CODE & GITHUB REPOSITORY

Key Backend Logic Snippets and Official Project Repository

DEFENSIVE SECURITY: PBKDF2 PASSWORD HASHING

```
from werkzeug.security import generate_password_hash, check_password_hash

# 1. Hashing password during registration
def register_user(username, email, raw_password):
    hashed_pw = generate_password_hash(
        raw_password, method='pbkdf2:sha256', salt_length=16
    )
    cursor.execute(
        "INSERT INTO users (username, email, password_hash) VALUES (%s, %s,
%s)",
        (username, email, hashed_pw)
    )
    db.commit()

# 2. Verifying password during login
if user and check_password_hash(user['password_hash'], input_password):
    session['user_id'] = user['id']
```

LENDING WORKFLOW: ATOMIC STATE TRANSITION

```
@app.route('/request/approve/<int:req_id>', methods=['POST'])
@login_required
def approve_request(req_id):
    # 1. Update borrow request status to Approved
    cursor.execute(
        "UPDATE borrow_requests SET status = 'Approved' WHERE id = %s",
        (req_id,)
    )
    # 2. Lock item status to 'Lent' to prevent double-booking
    cursor.execute(
        "UPDATE items SET status = 'Lent' WHERE id = %s",
        (request_item_id,)
    )
    db.commit()
    flash("Borrow request approved successfully!", "success")
    return redirect(url_for('dashboard'))
```

OFFICIAL PROJECT GITHUB REPOSITORY LOCATION

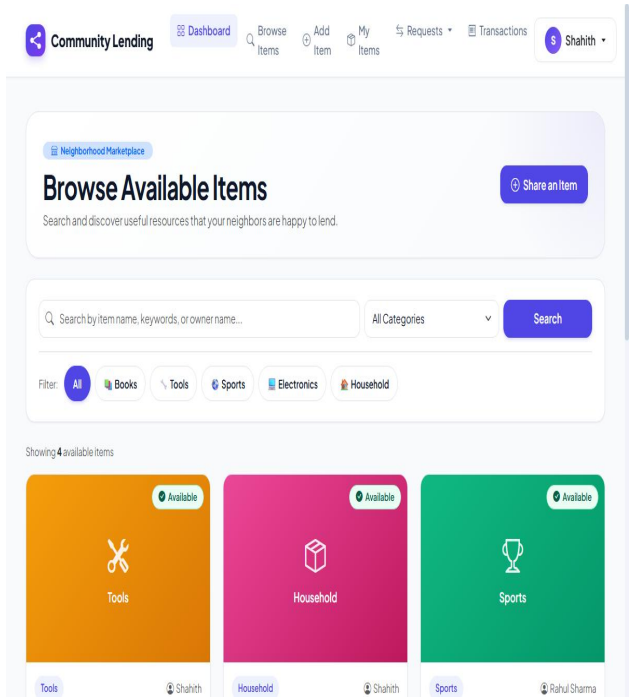
https://github.com/Shahith2004/Community-Item-Lending-System

Includes full Flask codebase, MySQL schema migrations, automated pytest suites, and documentation.

RESULTS AND APPLICATION INTERFACE

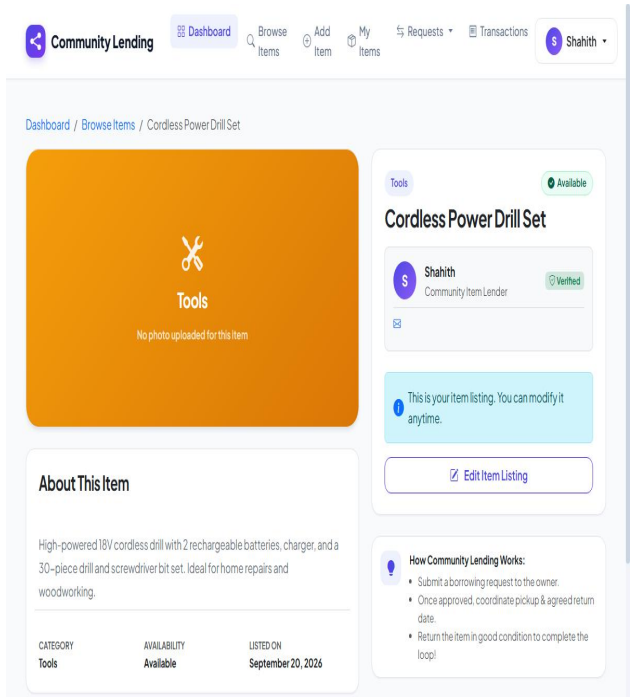
Live Production Screens of the Web Platform

Browse Community Items



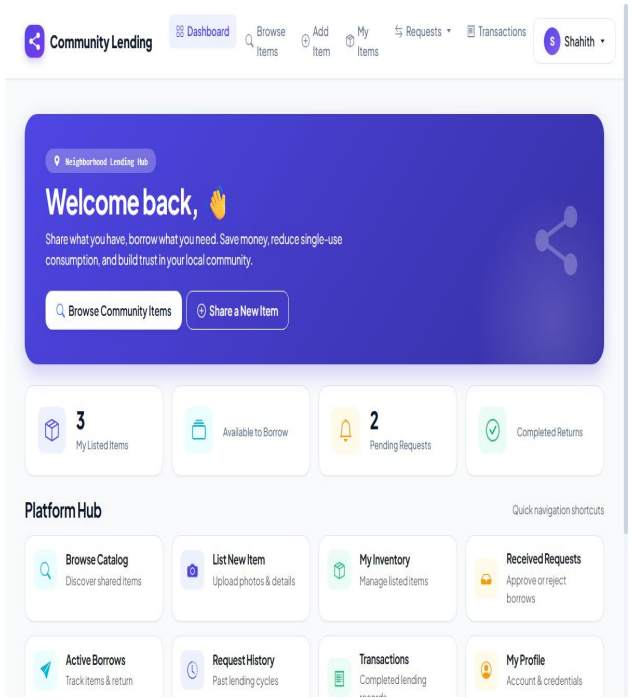
Search bar, category filters, and live availability cards.

Item Details & Request Modal



Item image, owner profile, and borrow submission form.

User Dashboard & Management



Unified view of listed items, requests, and active loans.

QUALITY ASSURANCE

TESTING & VALIDATION

V-Model Testing Strategy and Automated Pytest Results

PowerShell – Automated Test Execution: pytest (15 Test Cases)

```
PS C:\Users\Asus\Community-Item-Lending-System> pytest -v --disable-warnings
========================== test session starts ==========================
platform win32 -- Python 3.12.0, pytest-8.3.2, pluggy-1.5.0
rootdir: C:\Users\Asus\Community-Item-Lending-System
collected 15 items

tests/test_auth.py::test_tc01_user_registration                  PASSED  [  6%]
tests/test_auth.py::test_tc02_duplicate_email_rejected           PASSED  [ 13%]
tests/test_auth.py::test_tc03_valid_user_login                   PASSED  [ 20%]
tests/test_auth.py::test_tc04_invalid_password_rejected          PASSED  [ 26%]
tests/test_auth.py::test_tc05_protected_route_auth_guard         PASSED  [ 33%]
tests/test_catalog.py::test_tc06_add_item_listing                PASSED  [ 40%]
tests/test_catalog.py::test_tc07_disallowed_file_rejected        PASSED  [ 46%]
tests/test_catalog.py::test_tc08_keyword_search                  PASSED  [ 53%]
tests/test_catalog.py::test_tc09_category_filter                 PASSED  [ 60%]
tests/test_workflow.py::test_tc10_self_borrowing_prevented       PASSED  [ 66%]
tests/test_workflow.py::test_tc11_submit_borrow_request          PASSED  [ 73%]
tests/test_workflow.py::test_tc12_duplicate_request_prevented    PASSED  [ 80%]
tests/test_workflow.py::test_tc13_owner_approves_request         PASSED  [ 86%]
tests/test_workflow.py::test_tc14_owner_declines_request         PASSED  [ 93%]
tests/test_workflow.py::test_tc15_mark_item_returned             PASSED  [100%]

========================== 15 passed in 0.42s ==========================
```

TESTING METHODOLOGY & OUTCOMES

- **V-Model Testing Strategy:** Executed rigorous testing stages: Unit testing, Integration testing, and End-to-End System testing.
- **Pytest Automation Framework:** Developed an automated test suite using pytest and pytest-flask to validate endpoints and database queries.
- **Functional Coverage:** Tested user authentication, duplicate email prevention, item listing CRUD, UUID file uploads, and state machine transitions.
- **100% Pass Rate:** 15 comprehensive functional test cases executed; all 15 passed with 0 failures and zero regressions.
- **Defensive Boundary Checks:** Verified edge cases including unauthorized borrowing of own items and invalid file format rejection.

CONCLUSION & ACHIEVEMENTS

- **Operational Web Hub:** Delivered a complete, deployed platform for neighborhood item sharing.
- **Economic & Eco Impact:** Eliminated redundant buying and high rental shop charges.
- **Defensive Engineering:** PBKDF2 security, 4-stage state machine, and 100% test pass rate.

FUTURE SCOPE & ENHANCEMENTS

- **Cross-Platform Mobile Apps:** Develop iOS & Android native apps using Flutter or React Native.
- **Instant WhatsApp Alerts:** Push instant notifications via WhatsApp API when items are requested.
- **Reputation & Trust Badges:** 5-star community ratings and badges ('Top Lender', 'Punctual Returner').

THANK YOU!

Questions & Discussion

COMMUNITY ITEM LENDING SYSTEM

Candidate: MOHAMED SHAHITH AKEEL S A | Enrolment Number: EA2532251010500

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